

CCS356 - Object Oriented Software Engineering

Topic: use case model

1. SUMMARY

A use case model is a graphical representation of the interactions between a system and its users. It is a fundamental concept in object-oriented software engineering. The use case model describes the functional requirements of a system from the user's perspective. The key principles of a use case model include identifying the actors, defining the goals, and describing the interactions between the actors and the system. The reference material highlights that a use case model can be used to classify systems into different types, such as simple, complex, and chaotic. It also discusses the applications of use case models in software development, including requirement gathering, system design, and testing. The reference material emphasizes the importance of using use case models to ensure that the system meets the user's needs and expectations.

The use case model has several advantages, including improved communication between stakeholders, reduced misunderstandings, and increased customer satisfaction. However, it also has some disadvantages, such as increased complexity and the need for ongoing maintenance. The reference material provides examples of how use case models can be used in real-world applications, including e-commerce systems and healthcare systems.

2. SHORT NOTES

Detailed Definition with Examples

A use case model is a graphical representation of the interactions between a system and its users. It is a fundamental concept in object-oriented software engineering. The use case model describes the functional requirements of a system from the user's perspective. For example, in an e-commerce system, the use case

model would describe the interactions between the user and the system, including placing an order, paying for the order, and tracking the order.

Types and Classifications

There are several types and classifications of use case models, including:

- * Simple use case models: These are use case models that describe a single, straightforward interaction between the user and the system.
- * Complex use case models: These are use case models that describe multiple, complex interactions between the user and the system.
- * Chaotic use case models: These are use case models that describe unpredictable, dynamic interactions between the user and the system.

Step-by-Step Working Principle

The working principle of a use case model involves the following steps:

1. Identify the actors: The actors are the users or external systems that interact with the system.
2. Define the goals: The goals are the objectives that the actors want to achieve through the use case.
3. Describe the interactions: The interactions describe the sequence of events that occur between the actors and the system.
4. Identify the preconditions: The preconditions are the conditions that must be met before the use case can be executed.
5. Identify the postconditions: The postconditions are the conditions that are true after the use case has been executed.

Important Formulas and Equations

There are no specific formulas or equations associated with use case models.

Diagrams Description

A use case model can be represented using a variety of diagrams, including:

- * Use case diagram: This diagram shows the actors and the use cases that they interact with.
- * Sequence diagram: This diagram shows the sequence of interactions between the actors and the system.
- * State machine diagram: This diagram shows the states that the system can be in and the transitions between those states.

Advantages and Disadvantages

The advantages of use case models include:

- * Improved communication between stakeholders
- * Reduced misunderstandings
- * Increased customer satisfaction

The disadvantages of use case models include:

- * Increased complexity
- * Need for ongoing maintenance

Real-World Applications

Use case models can be used in a variety of real-world applications, including:

- * E-commerce systems
- * Healthcare systems
- * Financial systems

Comparison with Related Concepts

Use case models can be compared with other software engineering concepts, including:

- * Requirements gathering
- * System design
- * Testing

Time and Space Complexity

Use case models do not have time and space complexity.

Important Terminology

The following terminology is important in use case models:

- * Actor: The user or external system that interacts with the system.
- * Goal: The objective that the actor wants to achieve through the use case.
- * Interaction: The sequence of events that occur between the actor and the system.
- * Precondition: The condition that must be met before the use case can be executed.
- * Postcondition: The condition that is true after the use case has been executed.

3. PRACTICAL / CODING EXAMPLES

Since this subject is a hands-on implementation subject, we will provide some code examples.

C Implementation

```
```c
```

```
// Define the use case model structure
```

```
typedef struct {
 char *name;
 char *description;
 char *goal;
 char *precondition;
 char *postcondition;
} use_case;

// Function to add a use case to the model

void add_use_case(use_case **model, int *size) {
 use_case *new_use_case = malloc(sizeof(use_case));
 printf("Enter use case name: ");
 scanf("%s", new_use_case->name);
 printf("Enter use case description: ");
 scanf("%s", new_use_case->description);
 printf("Enter use case goal: ");
 scanf("%s", new_use_case->goal);
 printf("Enter use case precondition: ");
 scanf("%s", new_use_case->precondition);
 printf("Enter use case postcondition: ");
 scanf("%s", new_use_case->postcondition);
 model = realloc(model, sizeof(use_case) * (*size + 1));
 model[*size] = new_use_case;
 (*size)++;
}

// Function to print the use case model

void print_use_case_model(use_case *model, int size) {
 for (int i = 0; i < size; i++) {
 printf("Use Case %d:\n", i + 1);
 printf("Name: %s\n", model[i].name);
```

```
 printf("Description: %s\n", model[i].description);
 printf("Goal: %s\n", model[i].goal);
 printf("Precondition: %s\n", model[i].precondition);
 printf("Postcondition: %s\n\n", model[i].postcondition);
 }
}
...

```

### Python Implementation

```
```python
```

```
class UseCase:
```

```
    def __init__(self, name, description, goal, precondition, postcondition):
```

```
        self.name = name
```

```
        self.description = description
```

```
        self.goal = goal
```

```
        self.precondition = precondition
```

```
        self.postcondition = postcondition
```

```
class UseCaseModel:
```

```
    def __init__(self):
```

```
        self.use_cases = []
```

```
    def add_use_case(self):
```

```
        name = input("Enter use case name: ")
```

```
        description = input("Enter use case description: ")
```

```
        goal = input("Enter use case goal: ")
```

```
        precondition = input("Enter use case precondition: ")
```

```
        postcondition = input("Enter use case postcondition: ")
```

```
        use_case = UseCase(name, description, goal, precondition, postcondition)
```

```
        self.use_cases.append(use_case)
```

```
def print_use_case_model(self):
    for i, use_case in enumerate(self.use_cases):
        print(f"Use Case {i + 1}:")
        print(f"Name: {use_case.name}")
        print(f"Description: {use_case.description}")
        print(f"Goal: {use_case.goal}")
        print(f"Precondition: {use_case.precondition}")
        print(f"Postcondition: {use_case.postcondition}\n")

# Create a new use case model
model = UseCaseModel()

# Add use cases to the model
while True:
    print("1. Add use case")
    print("2. Print use case model")
    print("3. Exit")
    choice = input("Enter your choice: ")
    if choice == "1":
        model.add_use_case()
    elif choice == "2":
        model.print_use_case_model()
    elif choice == "3":
        break
    else:
        print("Invalid choice. Please try again.")
...

#### Algorithm
```

Study Material - Generated by AI

1. Define the use case model structure.
2. Add use cases to the model using the ``add_use_case`` function.
3. Print the use case model using the ``print_use_case_model`` function.

Time and Space Complexity

The time complexity of the ``add_use_case`` function is $O(1)$, as it only involves a constant amount of work. The space complexity is $O(n)$, as it involves allocating memory for the new use case.

The time complexity of the ``print_use_case_model`` function is $O(n)$, as it involves iterating over the use cases in the model. The space complexity is $O(1)$, as it only involves a constant amount of work.

4. IMPORTANT QUESTIONS

****Part-A (2 Marks)**** ? 5 questions, each with a crisp 2-3 line answer:

Q1. Define a use case model and list its key elements.

Answer: A use case model is a graphical representation of the system's functional requirements, consisting of actors, use cases, and relationships. Key elements include actors, use cases, associations, and extensions.

Q2. Compare use case model with data flow diagram.

Answer: Use case model focuses on system's functional requirements from user's perspective, whereas data flow diagram emphasizes on data processing and flow within the system. Use case model is more abstract and user-centric.

Q3. What is the purpose of a use case model in software development?

Answer: The primary purpose of a use case model is to capture the system's functional requirements from the user's perspective, facilitating effective communication among stakeholders and guiding the development process.

Q4. Write a short note on the importance of use case model in requirements gathering.

Answer: Use case model plays a crucial role in requirements gathering by providing a structured approach to identify, analyze, and document functional requirements, ensuring that the system meets user needs and expectations.

Q5. Provide an example of a use case model in a real-world application.

Answer: A use case model for an online shopping system might include actors such as customers and administrators, use cases like browsing products and processing payments, and relationships between them, illustrating the system's functional requirements.

****Part-B (13 Marks)**** ? 3 questions. Each answer MUST be at least 20-25 lines long.

Q1. Explain use case model in detail with neat diagram and suitable examples.

Answer:

A use case model is a graphical representation of the system's functional requirements, consisting of actors, use cases, and relationships. It provides a structured approach to identify, analyze, and document functional requirements from the user's perspective. The model consists of three main components: actors, use cases, and relationships. Actors represent external entities that interact with the system, such as users, devices, or other systems. Use cases represent the system's functional requirements, describing the interactions between actors and the system. Relationships between use cases and actors define the interactions and dependencies between them.

The use case model diagram typically consists of ovals representing use cases, stick figures representing actors, and lines representing relationships. For example, a use case model for a bank's online banking system might include actors such as customers and bank staff, use cases like account management and transaction processing, and relationships between them. The diagram would show the customers interacting with the account management use case, which in turn interacts with the transaction processing use case.

The working of a use case model involves the following steps:

1. Identify the actors: Determine the external entities that interact with the system.
2. Identify the use cases: Determine the system's functional requirements, describing the interactions

between actors and the system.

3. Define the relationships: Define the interactions and dependencies between use cases and actors.

4. Refine the model: Refine the model by adding more details, such as preconditions, postconditions, and extension points.

The use case model has several advantages, including improved communication among stakeholders, clearer understanding of system requirements, and better guidance for the development process. However, it also has some disadvantages, such as being time-consuming to create and maintain, and requiring significant expertise. Real-world applications of use case models include software development, business process modeling, and system engineering.

For instance, a use case model can be used to develop a software system for managing a university's student records. The model would include actors such as students, faculty, and administrators, use cases like student registration and course management, and relationships between them. The model would guide the development process, ensuring that the system meets the user needs and expectations.

In conclusion, the use case model is a powerful tool for capturing the system's functional requirements from the user's perspective, facilitating effective communication among stakeholders and guiding the development process. Its importance lies in its ability to provide a structured approach to requirements gathering, ensuring that the system meets user needs and expectations.

Q2. Describe the types of use case model with working principle and examples.

Answer:

There are several types of use case models, including:

1. **Essential Use Case Model**: Focuses on the system's essential functions, ignoring non-essential features.
2. **Concrete Use Case Model**: Includes all the details of the system, including non-essential features.
3. **Analysis Use Case Model**: Used during the analysis phase, focusing on the system's functional requirements.
4. **Design Use Case Model**: Used during the design phase, focusing on the system's architecture and components.

Each type of use case model has its own working principle and examples. For instance, the essential use case model works by identifying the system's essential functions and ignoring non-essential features. An example of an essential use case model is a use case model for a simple banking system, which would include use cases like account management and transaction processing, ignoring non-essential features like user authentication.

The concrete use case model works by including all the details of the system, including non-essential features. An example of a concrete use case model is a use case model for a complex banking system, which would include use cases like account management, transaction processing, and user authentication.

The analysis use case model works by focusing on the system's functional requirements during the analysis phase. An example of an analysis use case model is a use case model for a software system, which would include use cases like user registration and login, focusing on the system's functional requirements.

The design use case model works by focusing on the system's architecture and components during the design phase. An example of a design use case model is a use case model for a hardware system, which would include use cases like sensor data processing and actuator control, focusing on the system's architecture and components.

The following comparison table summarizes the different types of use case models:

Type	Focus	Example
Essential	Essential functions	Simple banking system
Concrete	All details	Complex banking system
Analysis	Functional requirements	Software system
Design	Architecture and components	Hardware system

In conclusion, each type of use case model has its own working principle and examples, and the choice of which type to use depends on the specific needs of the project.

Q3. Compare use case model with related concepts. Discuss advantages and applications.

Answer:

The use case model is related to other concepts such as data flow diagram, sequence diagram, and class diagram. Each of these concepts has its own advantages and applications.

The data flow diagram is a graphical representation of the system's data processing and flow, emphasizing the system's internal workings. The sequence diagram is a graphical representation of the system's interactions over time, emphasizing the system's dynamic behavior. The class diagram is a graphical representation of the system's classes and relationships, emphasizing the system's static structure.

The use case model has several advantages over these concepts, including:

- * Improved communication among stakeholders
- * Clearer understanding of system requirements
- * Better guidance for the development process

However, the use case model also has some disadvantages, such as:

- * Being time-consuming to create and maintain
- * Requiring significant expertise

The following comparison table summarizes the use case model and related concepts:

Concept	Focus	Advantages	Disadvantages
Use Case Model	System's functional requirements	Improved communication, clearer understanding, better guidance	Time-consuming, requires expertise
Data Flow Diagram	System's internal workings	Emphasizes data processing and flow	Ignores external interactions
Sequence Diagram	System's dynamic behavior	Emphasizes interactions over time	Ignores system's internal workings
Class Diagram	System's static structure	Emphasizes classes and relationships	Ignores system's dynamic behavior

In conclusion, the use case model is a powerful tool for capturing the system's functional requirements from the user's perspective, facilitating effective communication among stakeholders and guiding the development process. While it has some disadvantages, its advantages make it a widely used concept in software development and other fields.

****Part-C (15 Marks)**** ? 2 questions. Each answer MUST be at least 30 lines long.

Q1. Elaborate on use case model covering theory, implementation, real-world applications, and future scope.

Answer:

****Section 1: Deep theoretical background****

The use case model is based on the concept of functional requirements, which describe the system's behavior from the user's perspective. The model consists of actors, use cases, and relationships, which are used to capture the system's functional requirements. The use case model is a graphical representation of the system's functional requirements, providing a structured approach to identify, analyze, and document functional requirements.

The use case model has its roots in the concept of use cases, which were first introduced by Ivar Jacobson in the 1980s. The concept of use cases was later developed and refined by other researchers, leading to the development of the use case model. The use case model has been widely adopted in software development and other fields, becoming a standard tool for capturing functional requirements.

****Section 2: How it is implemented/used in practice****

The use case model is implemented in practice through a series of steps, including:

1. Identify the actors: Determine the external entities that interact with the system.
2. Identify the use cases: Determine the system's functional requirements, describing the interactions between actors and the system.
3. Define the relationships: Define the interactions and dependencies between use cases and actors.
4. Refine the model: Refine the model by adding more details, such as preconditions, postconditions, and extension points.

The use case model is used in practice in a variety of ways, including:

- * Software development: The use case model is used to capture the system's functional requirements, guiding the development process.
- * Business process modeling: The use case model is used to model business processes, identifying the key actors and use cases.
- * System engineering: The use case model is used to capture the system's functional requirements, guiding the design and development process.

****Section 3: Minimum 3 detailed real-world case studies or applications****

The use case model has been used in a variety of real-world applications, including:

- * Software development: The use case model was used to develop a software system for managing a university's student records. The model included actors such as students, faculty, and administrators, use cases like student registration and course management, and relationships between them.
- * Business process modeling: The use case model was used to model a company's sales process, identifying the key actors and use cases. The model included actors such as sales representatives and customers, use cases like sales order processing and inventory management, and relationships between them.
- * System engineering: The use case model was used to capture the functional requirements of a complex system, guiding the design and development process. The model included actors such as users and administrators, use cases like system configuration and monitoring, and relationships between them.

****Section 4: Limitations and challenges****

The use case model has several limitations and challenges, including:

- * Being time-consuming to create and maintain
- * Requiring significant expertise
- * Ignoring non-functional requirements
- * Focusing on functional requirements only

Despite these limitations, the use case model remains a widely used and effective tool for capturing functional requirements.

****Section 5: Future scope and improvements****

The use case model is continuously evolving, with new techniques and tools being developed to improve its

effectiveness. Some potential future developments include:

- * Integrating the use case model with other modeling techniques, such as data flow diagrams and sequence diagrams.
- * Developing new notations and tools to support the use case model.
- * Applying the use case model to new domains, such as artificial intelligence and machine learning.

In conclusion, the use case model is a powerful tool for capturing functional requirements, with a deep theoretical background, wide range of applications, and future scope for improvement.

Q2. Critically analyze use case model. Compare approaches, evaluate trade-offs, and justify with examples.

Answer:

****Section 1: Overview of the concept and its importance****

The use case model is a graphical representation of the system's functional requirements, consisting of actors, use cases, and relationships. It provides a structured approach to identify, analyze, and document functional requirements from the user's perspective. The use case model is an important tool in software development and other fields, facilitating effective communication among stakeholders and guiding the development process.

****Section 2: Different approaches/techniques used****

There are several approaches and techniques used to create a use case model, including:

- * ****Essential Use Case Model****: Focuses on the system's essential functions, ignoring non-essential features.
- * ****Concrete Use Case Model****: Includes all the details of the system, including non-essential features.
- * ****Analysis Use Case Model****: Used during the analysis phase, focusing on the system's functional requirements.
- * ****Design Use Case Model****: Used during the design phase, focusing on the system's architecture and components.

Each approach has its own strengths and weaknesses, and the choice of which approach to use depends on the specific needs of the project.

****Section 3: Detailed comparison of approaches with a table****

The following table compares the different approaches:

Approach	Focus	Advantages	Disadvantages
Essential	Essential functions	Improved communication, clearer understanding	Ignores non-essential features
Concrete	All details	Includes all features, comprehensive	Time-consuming, complex
Analysis	Functional requirements	Focuses on analysis phase, guides development	Ignores design and implementation
Design	Architecture and components	Focuses on design phase, guides implementation	Ignores analysis and functional requirements

****Section 4: Trade-off analysis (time, space, cost, complexity)****

The use case model involves several trade-offs, including:

- * ****Time****: Creating a use case model can be time-consuming, especially for complex systems.
- * ****Space****: The use case model requires significant space to represent the system's functional requirements.
- * ****Cost****: Creating a use case model can be costly, especially if it requires significant expertise.
- * ****Complexity****: The use case model can be complex, especially for large systems with many actors and use cases.

Despite these trade-offs, the use case model remains a widely used and effective tool for capturing functional requirements.

****Section 5: Justify the best approach with real examples****

The best approach to creating a use case model

5. MCQs

Q1. What is the purpose of a use case model in software engineering?

- a) To describe the internal workings of a system
- b) To describe the interactions between the system and its users
- c) To describe the hardware components of a system
- d) To describe the software components of a system

Answer: b) To describe the interactions between the system and its users

Explanation: A use case model is used to describe the interactions between the system and its users, including the goals, preconditions, and postconditions of the system.

Q2. What is the main advantage of using a use case model in software engineering?

- a) Improved communication between stakeholders
- b) Reduced complexity of the system
- c) Increased customer satisfaction
- d) Faster development time

Answer: a) Improved communication between stakeholders

Explanation: A use case model improves communication between stakeholders by providing a clear and concise description of the interactions between the system and its users.

Q3. What is the main disadvantage of using a use case model in software engineering?

- a) Increased complexity of the system
- b) Reduced customer satisfaction
- c) Slower development time
- d) Increased cost of development

Answer: a) Increased complexity of the system

Explanation: A use case model can increase the complexity of the system by requiring more detailed descriptions of the interactions between the system and its users.

Q4. What is the purpose of a precondition in a use case model?

- a) To describe the goal of the system
- b) To describe the preconditions of the system
- c) To describe the postconditions of the system
- d) To describe the interactions between the system and its users

Answer: b) To describe the preconditions of the system

Explanation: A precondition is a condition that must be met before the use case can be executed.

Q5. What is the purpose of a postcondition in a use case model?

- a) To describe the goal of the system
- b) To describe the preconditions of the system
- c) To describe the postconditions of the system
- d) To describe the interactions between the system and its users

Answer: c) To describe the postconditions of the system

Explanation: A postcondition is a condition that is true after the use case has been executed.

7. YOUTUBE LINKS

- [Use Case Model Tutorial](https://www.youtube.com/results?search_query=use+case+model+tutorial)
- [Use Case Model Example](https://www.youtube.com/results?search_query=use+case+model+example)

Study Material - Generated by AI

- [Use Case Model Best Practices](https://www.youtube.com/results?search_query=use+case+model+best+practices)
- [Use Case Model in Software Engineering](https://www.youtube.com/results?search_query=use+case+model+in+software+engineering)
- [Use Case Model for Beginners](https://www.youtube.com/results?search_query=use+case+model+for+beginners)